DB2HACK 2026 · ROUND 1 PROPOSAL · IBM DB2 + AGENTIC AI

Hotel Ops Copilot

A team of AI agents that runs a hotel's daily operations on **IBM Db2**. The AI only **proposes**; Db2 decides what it may read, **dry-runs** every action, enforces the business rules, and can **undo** any change.

> "The database governs the AI, not the prompt."

30/30
evaluation questions correct, checked against ground truth computed in Db2

24/24
red-team attacks blocked, 22 of them by Db2 itself

15+
Db2 capabilities doing real work: vectors, temporal tables, SQL PL, RCAC, EXPLAIN…

~4
LLM calls per question on average (30-question evaluation); a full morning brief takes ~16

Participant Clark Fernandes (solo) · GitHub clarkf19

Framework LangGraph + LangChain

Database IBM Db2 12.1 Community Edition

Status working prototype, 107 automated tests

THE PROBLEM

Problem statement

Every morning, a hotel duty manager must make revenue, guest and maintenance decisions from scattered data, and AI is too risky to hand the keys to

🧩 Fragmented insight

Occupancy lives in the property-management system, rates in a spreadsheet, and reviews across Google, TripAdvisor and OTAs. A rattling AC in room 307 is noticed only after **many** bad reviews, because nobody connects "aircon humming" and "AC noise" to one room.

📉 Slow, manual revenue decisions

Mid-size and independent hotels cannot afford revenue-management systems or analysts. Weak nights are discounted too late, sold-out conference nights are under-priced, and loyal VIPs arrive without anyone noticing it is their anniversary.

⚠️ AI is unsafe to deploy

"Chat with your database" tools can run any SQL, leak guest PII to a third-party model, follow instructions hidden in a review (prompt injection), and make changes nobody can audit or **undo**.

Who feels it

- **Duty and general managers** of independent and mid-size hotels: the morning brief, approvals and escalations
- **Revenue managers**: pricing soft and peak nights inside policy
- **Engineering and front office**: fixing the right room first, upgrading the right guest

The trust gap

No hotel will let an AI change prices or block rooms unless every change is **explainable, checked against business rules, approved by a human and reversible**. Prompts cannot guarantee that. **The database can.**

THE SOLUTION

Proposed solution

A governed multi-agent copilot: the AI proposes, a human approves, and Db2 enforces and can undo

The manager talks in plain English. LangGraph agents read Db2, find problems and opportunities, and queue evidence-backed actions. Every safety guarantee lives **inside IBM Db2**.

1 · Ask
"Give me this morning's brief" or any question in plain English

2 · Analyse
Router plus 3 specialist agents query Db2 (SQL, vectors, time travel)

3 · Propose
Actions go through `SP_PROPOSE_ACTION`, which validates them and blocks bad ones

4 · Dry run
Db2 executes the action in a transaction, shows its impact, then rolls back

5 · Approve
A human manager decides; `SP_APPLY_ACTION` re-checks the guardrails

6 · Undo
Any change is rolled back with `FOR SYSTEM_TIME AS OF`

Morning brief

One click: KPIs, guest issues and the actions to approve, composed from three agents' findings.

Complaint radar

Db2 vector search sorts every review sentence into themes and finds *where* each one clusters.

Dry run

See exactly what an approval would change, including oversold nights, before you approve.

Time travel

Every rate and booking version is kept; there's a change feed and one-click rollback.

Text-to-SQL analyst

Occupancy, ADR, RevPAR and pick-up from Db2 views, grounded in catalog comments.

Policy-grounded

The hotel handbook lives in Db2 via IBM `langchain-db2`; proposals cite policy IDs.

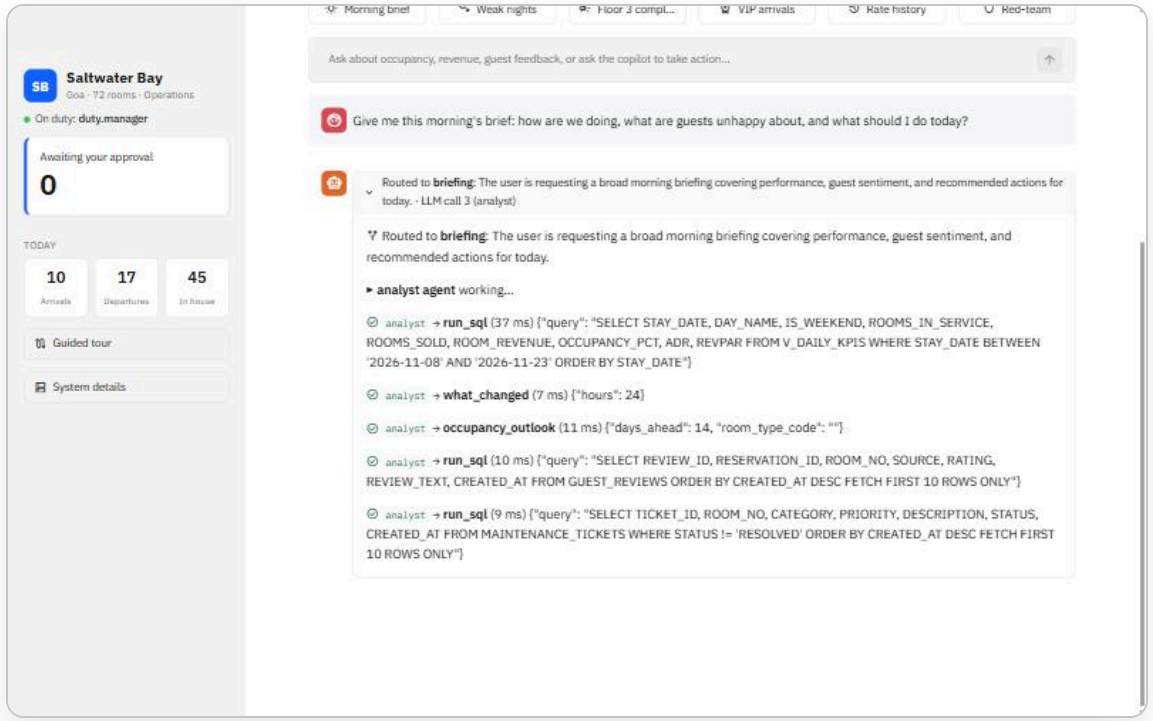
Privacy by Db2

Guest e-mail and phone are masked for the AI; opted-out guests are invisible to it (DPDP Act 2023).

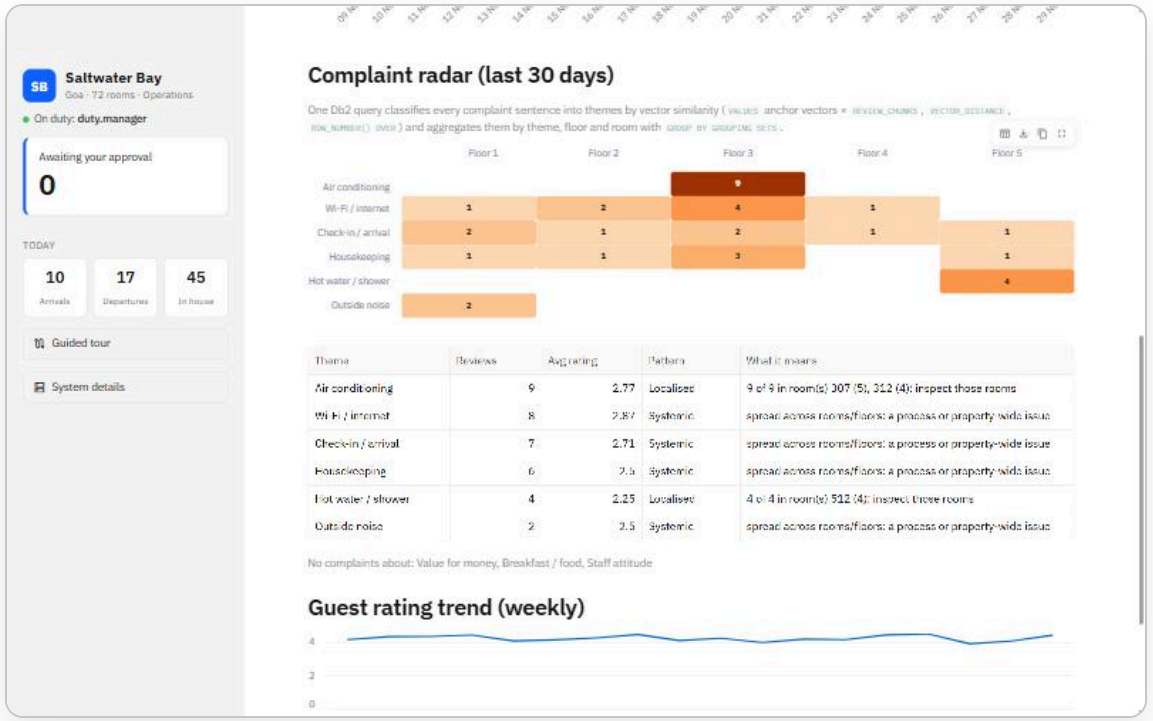
Earned autonomy

Managers grant the AI low-risk autonomy (e.g. ±5% price nudges). Db2 enforces the limits, and every change can be undone. 24/24 attacks are blocked.

From question to governed action in under a minute



Live multi-agent trace. The router picks the briefing route, then the analyst calls Db2 tools (SQL, change feed, outlook) with timings shown live. Every step is stored in Db2.



Complaint radar. One Db2 query classifies every complaint sentence. AC (rooms 307/312) and hot water (512) are *localised* faults; Wi-Fi and check-in are *systemic*.

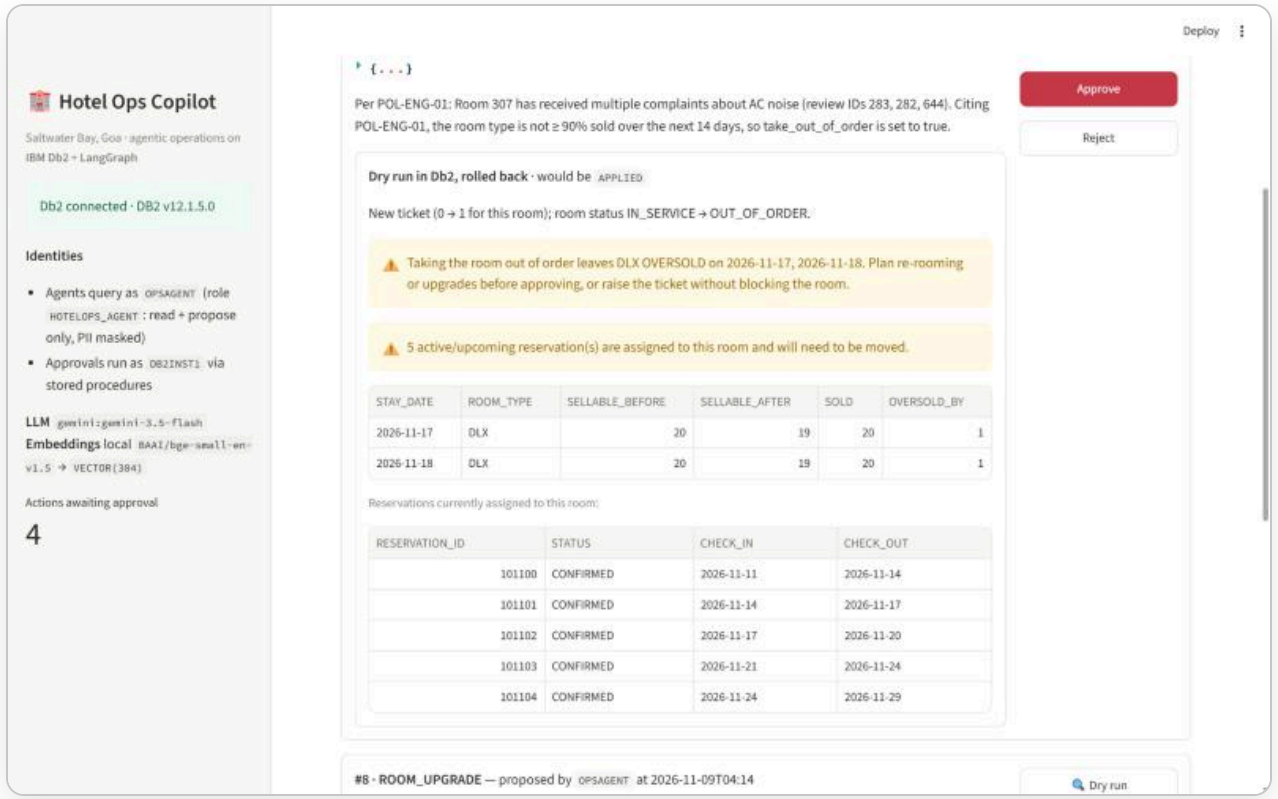
~16
LLM calls for a full morning brief across four agents (router, 3 specialists, composer)

12
Db2 tool calls in that brief: SQL, vectors, time travel and change feed

6
evidence-backed actions queued: two repairs, a VIP upgrade, three rate changes

100%
of steps stored in Db2: replayable, auditable and explainable

When the AI is wrong, Db2 catches it before any human approves



Real run on Gemini, not staged.

The operations agent cited policy `POL-ENG-01`, claimed Deluxe was "not ≥ 90% sold", and proposed taking AC-fault room 307 **out of order**.

The manager pressed **Dry run**. Db2 executed the real stored procedure inside a transaction, recomputed sellable inventory, and **rolled back**:

> ⚠️ Taking the room out of order leaves **DLX OVERSOLD on 17–18 Nov** (the conference nights). **5 reservations** are assigned to this room.

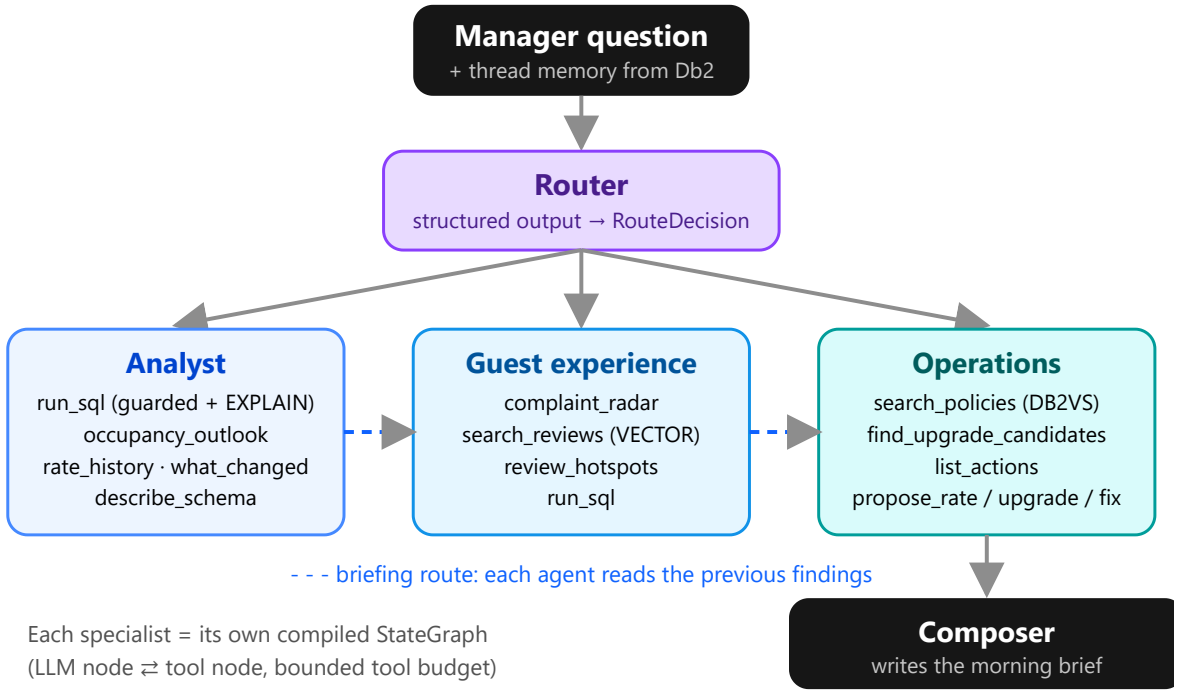
Why this matters

- Safety comes from **Db2** (procedures, transactions, privileges, RCAC), not from hoping the LLM behaves
- The AI can be fooled or wrong; the system stays correct
- Managers see **impact before approval** and can **undo after**

SELECTED AGENTIC AI FRAMEWORK

Agentic AI framework

LangGraph: explicit, auditable multi-agent control flow



Why LangGraph

- **Explicit graph**: deterministic routing, bounded loops and sub-graphs. Every step is inspectable and stored in Db2.
- **Least-privilege tools per agent**: the Analyst *cannot* propose actions, and only Operations can.
- **Multi-agent collaboration**: in a briefing, each specialist builds on the previous agents' findings.

Production-minded AI integration

- **Model**: Google Gemini 3.5 Flash through LangChain, swappable for watsonx.ai, OpenAI or Ollama
- **Quota-aware fallback chain** switches models mid-run, visible in the trace
- **Memory in Db2**: earlier turns are replayed from `AGENT_RUNS`
- **Untrusted data**: review text is never treated as instructions
- **Efficient**: batched tool calls, 3.4 LLM calls per answer on average

IBM Db2 connector

Native `ibm_db` plus IBM's official `langchain-db2`, with two Db2 identities

🤖 Agent identity

`OPSAGENT`, role `HOTELOPS_AGENT`: **SELECT** plus **EXECUTE SP_PROPOSE_ACTION** only. It sees PII masked and opted-out guests hidden.

👔 Manager identity

Approves, rejects, dry-runs and rolls back through stored procedures. Runs the EXPLAIN cost check.

CONNECTOR	USED FOR
`ibm_db` 3.3 (native driver)	All agent tools as LangChain `@tool`s; stored procedure calls with OUT parameters; prepared statements
IBM `langchain-db2` (`DB2VS`)	Policy handbook as a LangChain `VectorStore` on Db2 `VECTOR(384)`, through `ibm_db_dbi`
Local embeddings (BGE-small, ONNX)	Review and policy vectors, so text never leaves the machine to be embedded
Docker: Db2 Community Edition 12.1	`docker compose up` starts Db2 and the app together

```python
# every agent tool runs as the least-privileged identity
agent_db = get_db("agent")        # OPSAGENT, role HOTELOPS_AGENT

def run_sql(query: str) -> str:
    safe = validate_select(query)    # 1 statement, SELECT only
    check_cost(safe)                 # Db2 EXPLAIN PLAN, cost ceiling
    return agent_db.query(safe)       # Db2 privileges + RCAC apply

# agents can only PROPOSE: the procedure validates and queues
agent_db.call("HOTELOPS.SP_PROPOSE_ACTION", run_id, "RATE_CHANGE",
              json.dumps(payload), rationale, 0, OUT_TEXT)

# IBM langchain-db2: policy knowledge base on Db2 vectors
kb = DB2VS(embedding_function=LocalBgeEmbeddings(),
           table_name="POLICY_KB", client=ibm_db_dbi_conn,
           distance_strategy=DistanceStrategy.COSINE)
kb.similarity_search_with_score("upgrade a platinum guest")
```

The connection string sets `CURRENTSCHEMA=HOTELOPS`. Every credential comes from `.env`; none is committed.

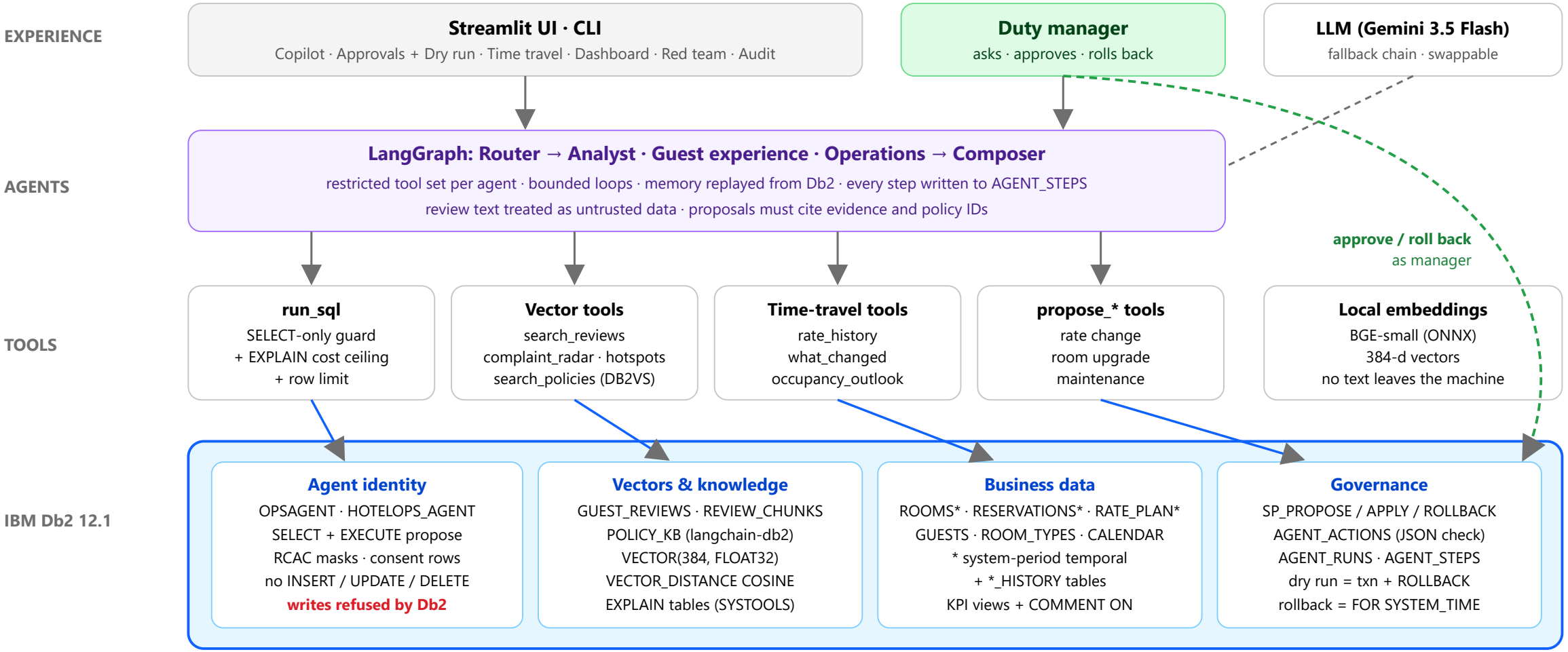
Intended Db2 integration

Db2 is the system of record *and* the AI's control plane

DB2 CAPABILITY	WHERE	WHAT IT DOES FOR THE COPILOT
VECTOR(384, FLOAT32) VECTOR_DISTANCE	`REVIEW_CHUNKS`, `POLICY_KB`	Sentence-level semantic review search; a complaint radar in one query (anchor vectors `CROSS JOIN`, `ROW_NUMBER() OVER`, `GROUPING SETS`)
System-period + bitemporal	`RATE_PLAN`, `RESERVATIONS`, `ROOMS`, `PROMOTIONS`	"Who changed the rate?"; a trigger-free change feed; one-click rollback (`FOR SYSTEM_TIME AS OF`); `BUSINESS_TIME` promotions answer "what was a guest quoted before the correction?"
SQL PL procedures	`SP_PROPOSE / APPLY / REJECT / ROLLBACK`	Price band, 35% cap, no past dates, real upgrades only, free room for the whole stay, consent, duplicates, and rollback integrity
Transactional dry run	dedicated unit of work + `ROLLBACK`	Runs the real apply procedure, diffs the impact (oversold nights, guests to move, revenue at stake), then persists nothing
RCAC masks + row permission	`GUESTS`	E-mail and phone masked for the AI; guests who withdrew consent are invisible to it (DPDP Act 2023)
Roles & GRANT / REVOKE	`HOTELOPS_AGENT`	The AI can read and propose, never write. `CREATETAB` and `IMPLICIT_SCHEMA` are revoked from PUBLIC
EXPLAIN PLAN	`SYSTOOLS.EXPLAIN_*`	The optimizer costs every AI-written query before it runs; runaway plans are refused (legitimate queries cost 78–2,089 timerons)
JSON_EXISTS / JSON_VALUE	`AGENT_ACTIONS`	Validated structured proposals (check constraint) parsed inside the procedures
Views, MQT, COMMENT ON, SYSCAT	`V_DAILY_KPIS`, `MQT_DAILY_KPIS`	A semantic layer for text-to-SQL; exact column lists are fed to the agents, so they don't guess
Tamper-evident audit	`AGENT_STEPS` trigger + `HASH`	Every tool call SHA-256 hash-chained by Db2 and anchored outside it: an edited step, or even a fully rebuilt ledger, is caught; replay without an LLM

System architecture

Four layers; the guarantees live in the bottom one



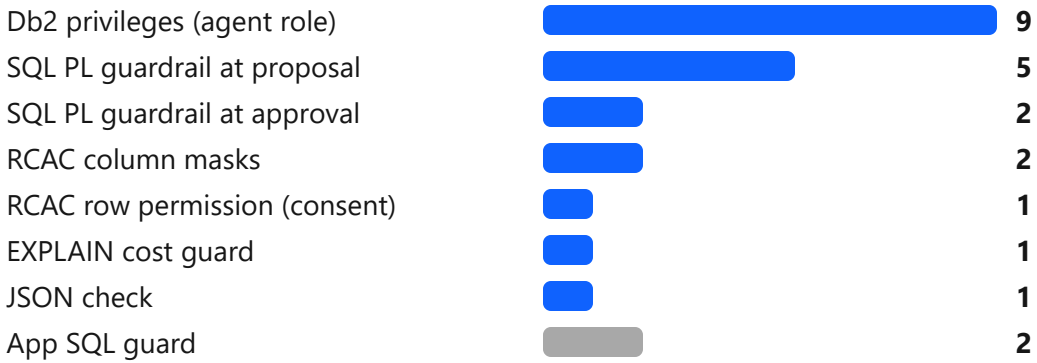
Workflow

Seven steps, five Db2 checkpoints

#	STEP	WHAT HAPPENS
1	**Question**	The manager asks; the run is logged in `AGENT_RUNS`, with memory replayed from earlier turns of the thread
2	**Routing**	The LLM router returns a typed `RouteDecision`, with a keyword fallback if structured output fails
3	**Evidence** Db2	Specialists query as `OPSAGENT`: SQL costed by EXPLAIN, vector search, time travel. Privileges and RCAC apply
4	**Proposal** Db2	`SP_PROPOSE_ACTION` validates the JSON, references, price band, dates, consent and duplicates, then queues it as PENDING
5	**Dry run** Db2	The real apply procedure runs in a transaction; the UI shows the diff and warnings, then `ROLLBACK`
6	**Approval** Db2	A human approves, or Db2 auto-applies the action if it is within the earned-autonomy limits (`AUTONOMY_POLICY`); `SP_APPLY_ACTION` re-checks the 35% cap, upgrade tier and room availability, then applies the change to the temporal tables
7	**Undo** Db2	`SP_ROLLBACK_ACTION` restores the rows `AS OF` the moment before the change, and refuses if someone changed them again

Red-team scoreboard: 24/24 attacks blocked

The attacks bypass the app and use the agent's own Db2 credentials. Each runs in a transaction that is rolled back.



Attacks include deleting guests, a ₹500 suite, self-approval, dumping PII, bypassing consent, a cross-join overload, a −40% price swing, a fake "upgrade", and **a prompt injection hidden in a guest review**.

FEASIBILITY: THIS IS ALREADY BUILT AND MEASURED

Feasibility: built and measured

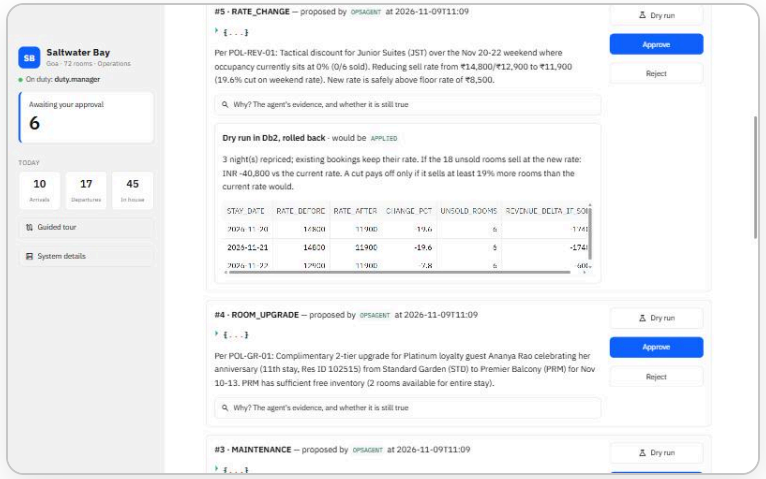
Measured results, not just intentions

30/30
accuracy evaluation: KPIs, time travel, vector search, change feed and safety, with ground truth computed in Db2

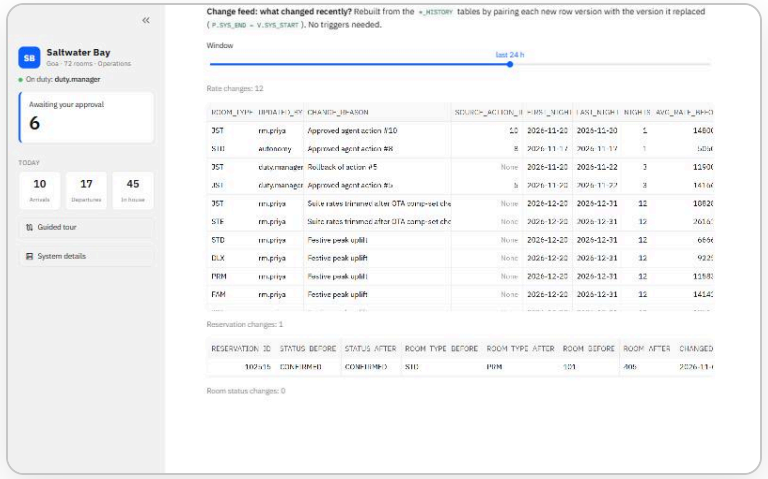
24/24
red-team attacks blocked; the database is proven unchanged afterwards

10,000
random proposals fuzzed into the Db2 guardrails: every decision matches an independent oracle

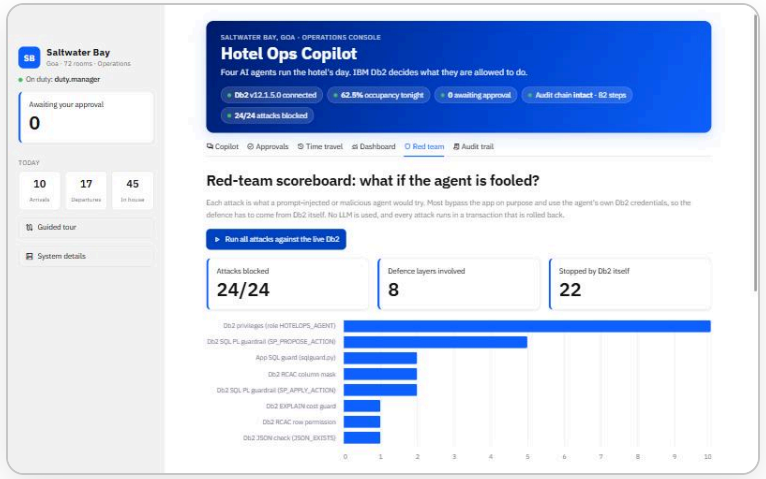
26/26
live Db2 checks of every claim (`verify.py` / Proofs tab); 107 tests, 87% coverage



Dry run shows the break-even: the agent's Junior Suite cut pays off only if it sells **19% more rooms**.



Change feed from history tables: before and after, who, why, and the action ID.



Red team: one click runs 24 attacks against the live Db2.

Better decisions every morning, without giving up control

💰 Revenue

- Soft nights are found days earlier and discounted **within policy**
- Sold-out conference nights are priced up, not left on the table
- The dry run shows money at stake and break-even before approval

⭐ Guest experience

- Faulty rooms are found from the **first few** reviews, not the fiftieth
- VIPs and special occasions are recognised automatically
- Process issues (Wi-Fi, check-in) are separated from room faults

🛡️ Trust & compliance

- The AI never writes data; every action is approved, audited and reversible
- PII never reaches the model; consent is enforced by Db2 (DPDP Act)
- Resistant to prompt injection by design

Use case: Monday, 07:30, Saltwater Bay (Goa)

The duty manager clicks **Morning brief**. Within minutes the copilot reports tonight's occupancy, flags AC faults in rooms 307/312 and no hot water in 512, spots zero suite pick-up for 20–23 Nov and the sold-out conference nights, and queues 6 actions: two repairs, an anniversary upgrade for a Platinum guest, and three rate changes. Each action can be dry-run first (in one recorded run this caught an action that would oversell Deluxe). The manager approves, and can undo any of them later.

Who can use it and how it scales

- **Independent and mid-size hotels** that lack revenue-management tools
- **Hotel chains**: the same Db2 row permissions can scope each manager and agent to their own property
- **Any regulated domain** where AI must propose and humans must approve: hospitals, retail, banking ops
- The domain is swappable; the **Db2 governance pattern is reusable**

IMPLEMENTATION PLAN, TECH STACK AND REPRODUCIBILITY

Reproducible in one command, built to be judged

Tech stack

- **Database**: IBM Db2 12.1 Community Edition (Docker)
- **Connectors**: `ibm_db`, IBM `langchain-db2`, `ibm_db_dbi`
- **Agents**: LangGraph + LangChain
- **LLM**: Gemini 3.5 Flash (fallback chain); watsonx.ai, OpenAI and Ollama ready
- **Embeddings**: BGE-small (fastembed, ONNX)
- **UI**: Streamlit + CLI; **tests**: pytest

Run it

```
# Db2 + app, no local Python
cp .env.example .env
docker compose up -d
# open http://localhost:8501

# or step by step
bash scripts/quickstart.sh
pytest -q tests/           # 107 tests
python src/redteam.py      # 24/24
python src/evaluate.py     # 30/30
```

It works without an API key too: approvals, dry run, time travel, radar, red team and replay of recorded runs.

Plan and roadmap

- ✅ Schema, temporal tables, vectors, SQL PL guardrails, RCAC
- ✅ LangGraph agents, tools, memory, fallback
- ✅ Dry run, radar, change feed, red team, evaluation, earned autonomy
- ✅ Docker, docs, screenshots
- → Next: PMS and channel-manager connectors, forecasting with Db2 ML / watsonx.ai, Granite as the default model, multi-property row permissions

Hotel Ops Copilot

AI that proposes. Db2 that decides, dry-runs and can undo.

Clark Fernandes · solo · GitHub **clarkf19**

IBM Db2 12.1 · LangGraph · Gemini